

Yiming Wang

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Research Profile

Hydrology and Earth-system data researcher with expertise in large-scale water mapping, multi-source satellite observations, hydrological data fusion, climate-impact analysis, machine learning, and high-performance scientific computing. Current Ph.D. research reconstructs global lake hydrological and thermal dynamics, quantifies compound hydro-climatic extremes, and develops scalable workflows for heterogeneous environmental datasets. Broader research interests include surface water dynamics, climate impacts on freshwater systems, hydrological extremes, and analysis of large environmental datasets.

Education

Tsinghua University, Beijing, China

Ph.D. in Hydraulic Engineering, Department of Hydraulic Engineering Sep 2021–Present
Expected completion: Jun 2026

Dissertation: *Hydrological and Thermal Dynamics of Global Lakes and Their Impacts on Littoral Zones and Thermal Habitats*

Advisor: Prof. Di Long

Tsinghua University, Beijing, China

B.Eng. in Hydraulic Engineering Sep 2017–Jun 2021

Thesis: *Monitoring of Water Surface Extent in China's Lakes and Reservoirs Using Optical and SAR Satellites*

Advisor: Prof. Di Long

Tsinghua University, Beijing, China

B.A. in Economics (Second Major), School of Economics and Management Sep 2018–Jun 2021

Thesis: *The Impact of Abolishing Agricultural Tax on Grain Yield*

Selected Publications

* *Corresponding author*

1. **Wang, Y.**, Long, D.*, & Li, X. (2023). High-temporal-resolution monitoring of reservoir water storage of the Lancang-Mekong River. *Remote Sensing of Environment*, 292, 113575. [doi:10.1016/j.rse.2023.113575](https://doi.org/10.1016/j.rse.2023.113575).
2. **Wang, Y.**, Li, C., Cui, Y., Cui, Y., Xu, Y., Hora, T., Zaveri, E., Rodella, A.-S., Bai, L., & Long, D.* (2024). Spatial downscaling of GRACE-derived groundwater storage changes across diverse climates and human interventions with Random Forests. *Journal of Hydrology*, 640, 131708. [doi:10.1016/j.jhydrol.2024.131708](https://doi.org/10.1016/j.jhydrol.2024.131708).
3. **Wang, Y.**, Long, D.*, Li, X., Li, L., Scanlon, B. R., & Woolway, R. I. (2026). Global compression and reorganization of lake thermal habitats. Under review at *Nature Geoscience*.
4. Li, L., Long, D.*, **Wang, Y.**, & Woolway, R. I. (2025). Global dominance of seasonality in shaping lake-surface-extent dynamics. *Nature*, 642, 361–368. [doi:10.1038/s41586-025-09046-3](https://doi.org/10.1038/s41586-025-09046-3).
5. Chen, Y., **Wang, Y.**, Li, L., Cui, Y., Duan, X., & Long, D.* (2024). Monthly monitoring of inundated areas and water storage dynamics in China's large reservoirs using multisource remote

sensing. *Water Resources Research*, 60, e2023WR036450. doi:10.1029/2023WR036450.

6. Xu, Y., Long, D. *, Li, X., **Wang, Y.**, Zhao, F., & Cui, Y. (2024). Unveiling lake ice phenology in Central Asia under climate change with MODIS data and a two-step classification approach. *Remote Sensing of Environment*, 301, 113955. doi:10.1016/j.rse.2023.113955.
7. Zhao, F., Long, D. *, Fang, C., **Wang, Y.**, & Duan, X. (2026). Satellite-observed surging dynamics of North Kunchang Glacier I in the Eastern Karakoram. *The Cryosphere*, 20, 2143–2167. doi:10.5194/tc-20-2143-2026.
8. Dong, L., Long, D. *, Zhang, C., Cui, Y., Cui, Y., **Wang, Y.**, Li, L., Hong, Z., Yao, L., Quan, J., Bai, L., Wang, H., & Scanlon, B. R. (2024). Shifting agricultural land use and its unintended water consumption in the North China Plain. *Science Bulletin*, 69, 3968–3977. doi:10.1016/j.scib.2024.11.009.
9. Hong, Z., Long, D. *, Li, X., **Wang, Y.**, Zhang, J., Hamouda, M. A., & Mohamed, M. M. (2023). A global daily gap-filled chlorophyll-*a* dataset in open oceans during 2001–2021 from multisource information using convolutional neural networks. *Earth System Science Data*, 15, 5281–5300. doi:10.5194/essd-15-5281-2023.

Selected Conference Presentations

- **Wang, Y.**, Long, D., & Li, X. (2022). High-temporal-resolution monitoring of reservoir water storage in the transboundary Lancang-Mekong River Basin in Southeast Asia. *AGU Fall Meeting 2022*, Chicago, IL, USA. Oral presentation.
- **Wang, Y.**, Li, C., Long, D., Hora, T., Zaveri, E., & Rodella, A.-S. (2023). Downscaling GRACE satellite-derived groundwater storage changes over sub-Saharan Africa, the Middle East, and South Asia using Random Forests. *AGU Fall Meeting 2023*, San Francisco, CA, USA. Poster presentation.
- **Wang, Y.**, Long, D., Li, X., Li, L., Scanlon, B. R., & Woolway, R. I. (2024). Investigating changes in lake littoral habitats under climate change. *AGU Fall Meeting 2024*, Washington, D.C., USA. Poster presentation.
- **Wang, Y.**, Long, D., & Li, L. (2025). Leveraging deep learning and thermodynamic models for projecting hydro-thermal dynamics of lakes on the Tibetan Plateau. *AGU Annual Meeting 2025*, New Orleans, LA, USA. Poster presentation.

Research Experience

Graduate Research Assistant, Tsinghua University, Beijing, China Sep 2021–Present

Project: Multi-source water mapping, storage estimation, and climate-impact analysis

- Integrated optical, SAR, laser-altimetry, DEM, climate, and hydrological datasets to estimate surface-water extent, water level, and storage dynamics at basin to global scales ([Publication 1](#); [4](#); [5](#)).
- Built and validated high-temporal-resolution reservoir storage estimates for the transboundary Lancang–Mekong River, including uncertainty assessment from non-uniform satellite observations and topographic constraints.
- Reconstructed global lake hydrological and thermal dynamics for more than 1.4 million lakes and analysed climate-driven low-water and high-temperature compound extremes ([Publication 3](#); [4](#)).
- Developed reproducible geospatial workflows for large-scale processing, model-data comparison, validation, and publication-quality visualisation on high-performance computing systems.

Project: Groundwater storage variations and climate-human drivers

- Applied Random Forest models to downscale GRACE-derived groundwater storage changes across regions with contrasting climates and human water use ([Publication 2](#)).

- Integrated climate, hydrology, land-use, and water-management predictors to generate high-resolution groundwater-change maps.
- Interpreted model outputs in terms of regional climate variability, anthropogenic pressure, and water-security implications.

Teaching Experience

Graduate Teaching Assistant, Tsinghua University 2021–2024
Theories and Applications of Remote Sensing

- Delivered tutorials on remote sensing data access, acquisition workflows, image preprocessing, enhancement, and classification.
- Guided students in supervised and unsupervised classification using ENVI.

Graduate Teaching Assistant, Tsinghua University 2022–2023
Remote Sensing Hydrology

- Assisted with advanced satellite remote sensing instruction and student feedback.

Awards and Honors

- **Zhang Guangdou Science and Technology Education Fund Scholarship**, 2020 and 2024. Awarded to outstanding students in Hydraulic Engineering in China.
- **Wiley China Excellent Author Program**, Jul–Sep 2024.
- **Special Prize, Tsinghua University Challenge Cup**, 2020. Awarded to the top 10 university-wide student academic science and technology projects.

Core Strengths and Technical Skills

Strengths: large-scale hydrological data integration; water-surface and storage mapping; hydroclimatic and climate-impact analysis; uncertainty-aware model-data comparison; scalable geospatial workflows.

Research fit: large-scale environmental data analysis; surface-water and storage dynamics; flood-, drought-, and groundwater-relevant hydrological variability; climate-sensitive freshwater systems; remote-sensing constraints for hydrological inference.

Programming: Python, MATLAB, R, Fortran

Tools and Platforms: HPC / supercomputing, Google Earth Engine, ArcGIS, ENVI

Methods: machine learning, Random Forests, multimodal remote-sensing fusion, thermodynamic lake modelling, uncertainty-aware validation, geospatial analysis

Languages: Chinese (native), English (professional working proficiency)